Exam Template — Course X

Department of Physics — University Y
Course: General Topic
Semester 2026-1

Name: ______________________ **ID Number:** ________ **Programme:** ________

---

1. **Generic Problem in Relativistic Mechanics (30 pt).**

   State the general context of the problem here.

   a) Derive a symbolic expression for a physical quantity of interest.

      Solution:

      This is a sample solution. Replace this text with the actual derivation.

      Discuss the physical interpretation of the previous result and its limiting cases.

      Solution:

      Include a brief interpretation and a discussion of limiting cases here.

Exam Template — Course X

Department of Physics — University Y
Course: General Topic
Semester 2026-1

Name: ______________________ **ID Number:** ________ **Programme:** ________

---

2. **Generic Problem in Electromagnetic Fields (40 pt).**
   Consider a scenario with fields $\vec{E}$ and $\vec{B}$ under a Lorentz transformation.
   a) Determine the fields in a moving reference frame and compare with the original frame.
      Solution:
      Write the transformation equations and a qualitative comparison of results here.

3. **Single-Answer and Multiple-Answer Questions (30 pt).**
   Select the correct option(s) and provide a brief justification.
   a) **(Single Choice)** For a free particle, the conserved quantity associated with time-translation symmetry is:
      a) The total angular momentum.
      b) The total energy.*
      c) The effective electric charge.
      d) The local scalar curvature.
   b) **(Multiple Choice)** In special relativity, which of the following statements are correct?
      a) The speed of light in vacuum is invariant.*
      b) Proper time is the same for all inertial observers.
      c) Events simultaneous in one frame may not be simultaneous in another.*
      d) Rest mass depends on the reference frame.

4. **Short Conceptual Questions (30 pt).**
   Answer briefly and with justification.
   a) **General Concepts** Describe in two or three sentences the difference between invariant quantities and observer-dependent quantities.
      Solution:
      An invariant keeps its value under Lorentz transformations, whereas an observer-dependent quantity changes with the reference frame.
   b) **Application** Propose an example where the choice of reference frame simplifies the analysis of a problem.
      Solution:
      A typical example is working in the centre-of-mass frame for relativistic collisions, which simplifies the conservation of energy and momentum.

Formula Sheet

- Euler–Lagrange equations: $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$
- Lagrange–Jacobi identity and Virial theorem: $\dot{G} = 2K + U$, $2\langle K\rangle = -\langle U\rangle$
- Lorentz transformations (1D): $x' = \gamma(x - vt), \quad t' = \gamma\left(t - \frac{vx}{c^2}\right)$